

Brindlewythe Biosciences Ltd, a Tarnovell company

TP04 kinase selectivity reassessment after scaffold substitution

Final study report | Revision B

Brindlewythe Biosciences Ltd, a Tarnovell company

Study BW21_0240

Accession B-ACC-2021-04086

Client: Tervemere Pharma AG

Date: 19 Aug 2021

Author: Dr B. Rowcroft, Screening Operations Director

1 Summary and plate acceptance

24 TP04 compounds were tested in the Primary potency assay. The best IC₅₀ was 0.55 nM. The Selectivity sweep covered the representatives named in the panel table at 1 µM.

Primary and panel plates were accepted before interpretation. Vehicle and background responses were assessed alongside reference inhibitor concentration responses. The primary endpoint and panel endpoint are released separately.

Table 1. Study benchmark

Metric	Value
Project	BW21_0240
Client	Tervemere Pharma AG
Compounds tested	24
Series / chemotype	Tervemere aminopyrazolopyrimidine series 04 (Aminopyrazolopyrimidine)
Assays	Primary kinase IC ₅₀ , Kinase selectivity panel
Turnaround (calendar days)	45
Price	GBP 18,400

Control acceptance precedes interpretation. The benchmark summarises the study; the complete compound tables below are the basis of selection. Screening decisions are recorded separately from the endpoints.

2 Principal results

Table 2. Primary potency assay: IC50 and Hill slope

Compound	IC50 (nM)	Hill slope
TP04-02706	8.2	1.22
TP04-02716	0.55	1.20
TP04-02724	34	1.11
TP04-02728	73	0.89
TP04-02734	18	0.94
TP04-02748	1.8	0.92
TP04-02764	1.3	0.92
TP04-02778	10.0	1.07
TP04-02782	6.6	0.95
TP04-02784	3.8	1.03
TP04-02800	32	1.03
TP04-02806	1.3	1.10
TP04-02814	17	0.87
TP04-02824	25	0.99
TP04-02826	9.8	1.08
TP04-02830	3.7	1.01
TP04-02836	20	0.91
TP04-02838	44	0.99
TP04-02850	15	1.02
TP04-02862	8.3	1.20
TP04-02868	5.1	0.98
TP04-02872	55	0.90
TP04-02874	7.7	1.23
TP04-02880	109	0.96

ATP at Km; 10-point, 3-fold; staurosporine control

Table 3. Selectivity sweep: percent inhibition at 1 μ M

Kinase	TP04-02706	TP04-02716	TP04-02724	TP04-02728	TP04-02734	TP04-02748
ABL1	55	12	0	26	38	40
AURKA	69	34	0	24	21	28
CDK2	49	31	0	12	33	37
CHEK1	12	52	17	40	25	31
EGFR	42	15	7	18	58	36
FLT3	41	36	11	51	5	19
GSK3B	36	17	18	43	3	0
JAK2	31	10	28	36	46	24
KDR	62	59	11	19	51	50
LCK	18	39	16	46	19	0
MAPK14	85	45	26	74	44	79
PIK3CA	20	23	58	74	66	18
SRC	0	43	20	6	36	38
SYK	26	44	56	60	17	45
BTK	54	25	19	0	0	8
ROCK1	24	45	51	36	19	40

Compound identifiers follow the sample cards. Units are part of the headings. Compare like endpoints under the stated conditions. A position in a table is not an estimate of clinical performance.

3 Study remit and chronology

Tervemere Pharma AG commissioned Primary potency assay and Selectivity sweep to support a chemistry screen gate for this screening set. The remit was not a complete disposition or safety assessment.

Study work began on 05 Jul 2021. The interim checkpoint was 27 Jul 2021. This final revision closes the study on 19 Aug 2021. Dr B. Rowcroft is responsible for scientific release; Ines Alvarado checked plate acceptance and table transfer.

The completed study covers 24 compounds under accession B-ACC-2021-04086. The accession links submitted material, sample cards and run sheets. No later screening set is included.

4 Test-lot handling

TP04 sample-card identifiers were retained in selection of sweep representatives. The panel columns define its scope; they must not be read as if every member of the primary screening set had been tested against each off-target kinase.

Sample-card identifiers carried through preparation, assay allocation and result collation. Working solutions were prepared for the scheduled assay with matched vehicle. Results refer to the submitted test lot, rather than a claim that a later resynthesis will behave identically.

Preparation details, positions and handling observations remain in the run sheets. Qualifications stay with the endpoint; they are not removed to simplify a ranking.

5 Assay procedure

The Primary potency assay used recombinant inflammatory-signal kinase with ATP at Km. A 10-point, threefold concentration dilution was tested with matched vehicle content. Staurosporine reference wells and background wells accompanied the compound responses.

Concentration responses were fitted to obtain IC50 and Hill slope. Fits were considered with response span and reference behaviour. Potency is reported in nM; ATP conditions remain part of the interpretation.

The Selectivity sweep measured percent inhibition at 1 μ M against the off-target kinases listed in the table. Representatives retained their TP04 identifiers across both assays. These single-concentration results were not fitted as off-target IC50 values.

6 Control assessment and release

Primary and panel plates were accepted before interpretation. Vehicle and background responses were assessed alongside reference inhibitor concentration responses. The primary endpoint and panel endpoint are released separately.

Reference inhibitor behaviour supported release of each assay. Ines Alvarado checked plate acceptance and table transfer. Numerical control values are retained in the run sheets; no unprovided Z-prime or reference IC50 is added to the report.

Ines Alvarado checked that control assessment preceded the screening interpretation. Dr B. Rowcroft authorised release. Acceptance supports use within the stated assay; it does not eliminate compound-specific questions that need follow-up.

7 Data reduction

Primary concentration responses yield IC₅₀ and Hill slope. Panel results are vehicle-normalised percent inhibition at 1 μ M. Primary potency and the off-target pattern answer different questions even when the compound identifier is shared.

Reported precision and units are retained. Rounding must not manufacture distinctions between near neighbours. The screening decision should use complete rows with qualifications visible.

Summary endpoints and individual observations have different purposes. The benchmark does not replace the detailed table. No additional fitted values or composite scores have been inferred for presentation.

8 Screening-set interpretation

The revised hinge core reduced off-target activity in Tervemere's screening assessment. The panel should be read by compound and kinase, rather than reduced to primary potency alone. Retaining polar substituents is a chemistry decision; it does not establish a binding mode.

The primary table identifies potent members of the revised screening set; the sweep describes selected representatives away from the intended target. The scaffold comparison remains qualitative here because an earlier screening-set dataset is not appended.

A chemistry association can identify the next experiment without establishing the molecular cause of the assay response. This report makes no target-mechanism claim from screening data alone.

9 Screen gate

The revised hinge core reduced off-target activity and Tervemere retained the polar substituents for the next design cycle.

Retain the polar substituents in the next design cycle. Where an off-target response would alter selection, obtain a concentration response before assigning a quantitative margin. New designs require their own measurements.

The screen gate records the client's use of the data. A proposed follow-up is not a completed result. No endpoint from that later work is incorporated into this signed report.

10 Study limitations

Biochemical potency depends on the assay conditions, including ATP. It is not cellular efficacy or target engagement in tissue.

The sweep covers the listed kinases and representatives. It does not establish family-wide selectivity or panel results for every primary-screen compound.

Single-concentration inhibition cannot provide a selectivity ratio based on off-target IC₅₀. A flag identifies a follow-up question, not an affinity measurement.

These endpoints apply to the submitted screening set and test lots. No extrapolation is made to an untested analogue, formulation or biological system. Later measurements should retain identifiers and conditions for comparison.

11 Conclusions

Primary potency assay and Selectivity sweep is complete for BW21_0240. Control assessment supports release of the detailed tables for the specified client decision.

The revised hinge core reduced off-target activity and Tervemere retained the polar substituents for the next design cycle.

The results support the stated next step. They do not alone resolve exposure, efficacy or nomination suitability. Any later integrated assessment must identify its additional supporting studies.

12 Record index and authorisation

The retained study record comprises sample cards linked to B-ACC-2021-04086, preparation and assay run sheets, control assessment, result tables and this final report. The interim checkpoint forms part of the chronology and does not replace the completed dataset.

Revision B is the final report issued to Tervemere Pharma AG on 19 Aug 2021. Read principal results with the methods and limitations. Preserve study identity, units and qualifications when using tables for the screen gate.

Scientific release: Dr B. Rowcroft, Screening Operations Director. Plate acceptance and report-table transfer: Ines Alvarado, Assay Quality Coordinator. Signatures apply to this completed study and its bounded interpretation.

Name	Role	Signature	Date
Dr B. Rowcroft	Screening Operations Director — scientific release	<u>D Rowcroft</u>	19 Aug 2021
Ines Alvarado	Assay Quality Coordinator — plate acceptance and table transfer	<u>I Alvarado</u>	19 Aug 2021